CTL DECISION SHEET

Standard(s):	Sub-clauses:	No.	Year
IEC 60695-2-11 **IEC 60695-2-12** **IEC 60695-2-13**	**11 - 12**	**DSH 0752**	**2009**
Category: **VARIOUS - GENERAL**		**Developed by:** **WG4 – WG2**	
Subject: **Definition of a flame for glow wire test**	**Key words:** **- Glow wire test** **- Flame** **- Corona Effect** **- Ionized gasses** **- Ignition of the sample**	Approved at the 47th CTL Plenary Meeting, year 2010.	

Question:

How is the “Corona-Effect” to be interpreted when performing the Glow Wire Test?

Decision:

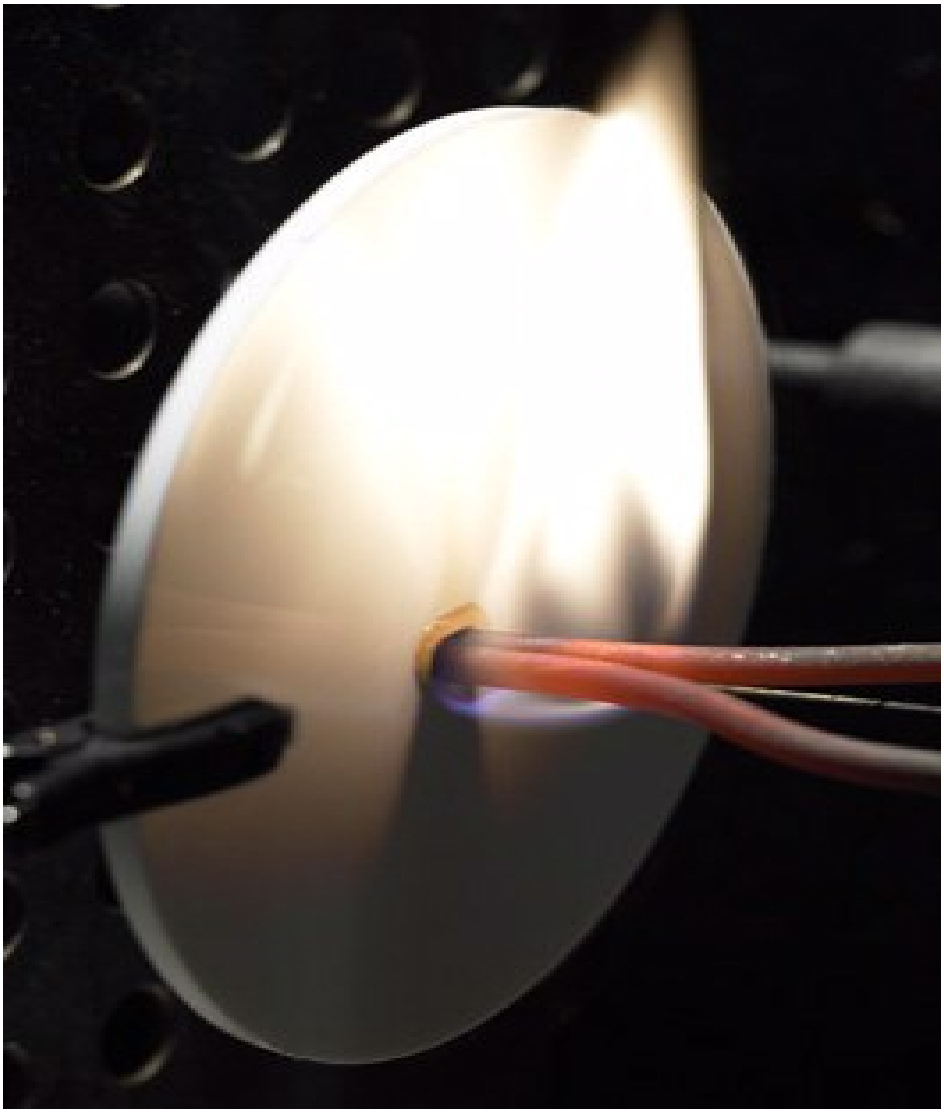
During the glow wire test (according to the temperature settings required by the standard), the following phenomena can occur:

1. If a brightly shining flame is observed directly at the test specimen (see picture 1), this is a clear indication of an ignition. Some materials show a colourless transparent flame instead of a bright shiny flame. This is considered to be ignition of the test specimen, if the maximum flaming time allowed by the standard is exceeded.

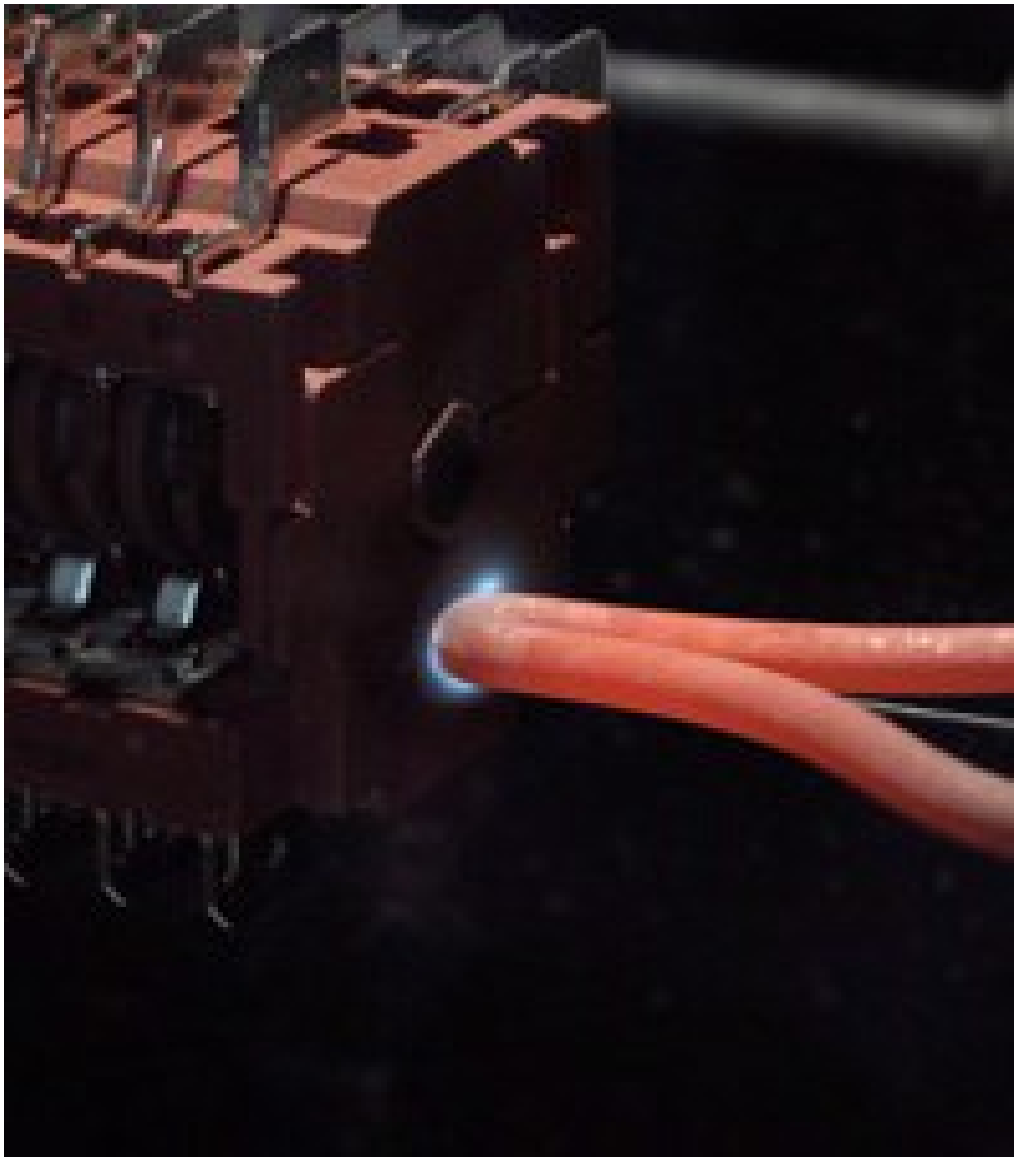
2. Sometimes, a circular corona can be observed around the tip of the glow wire, see picture 2. This corona consists of ionized gas, predominantly blue in colour and is located in the immediate vicinity of the glow wire. This phenomenon is not considered to be ignition of the test specimen, as long as it does not lead to an ignition of the test specimen according to point 1.

3. Furthermore, ionized gasses can sometimes be noticed along the glow wire (see picture 3). This effect is also not considered to be ignition of the test specimen, as long as it does not lead to an ignition of the test specimen according to point 1.

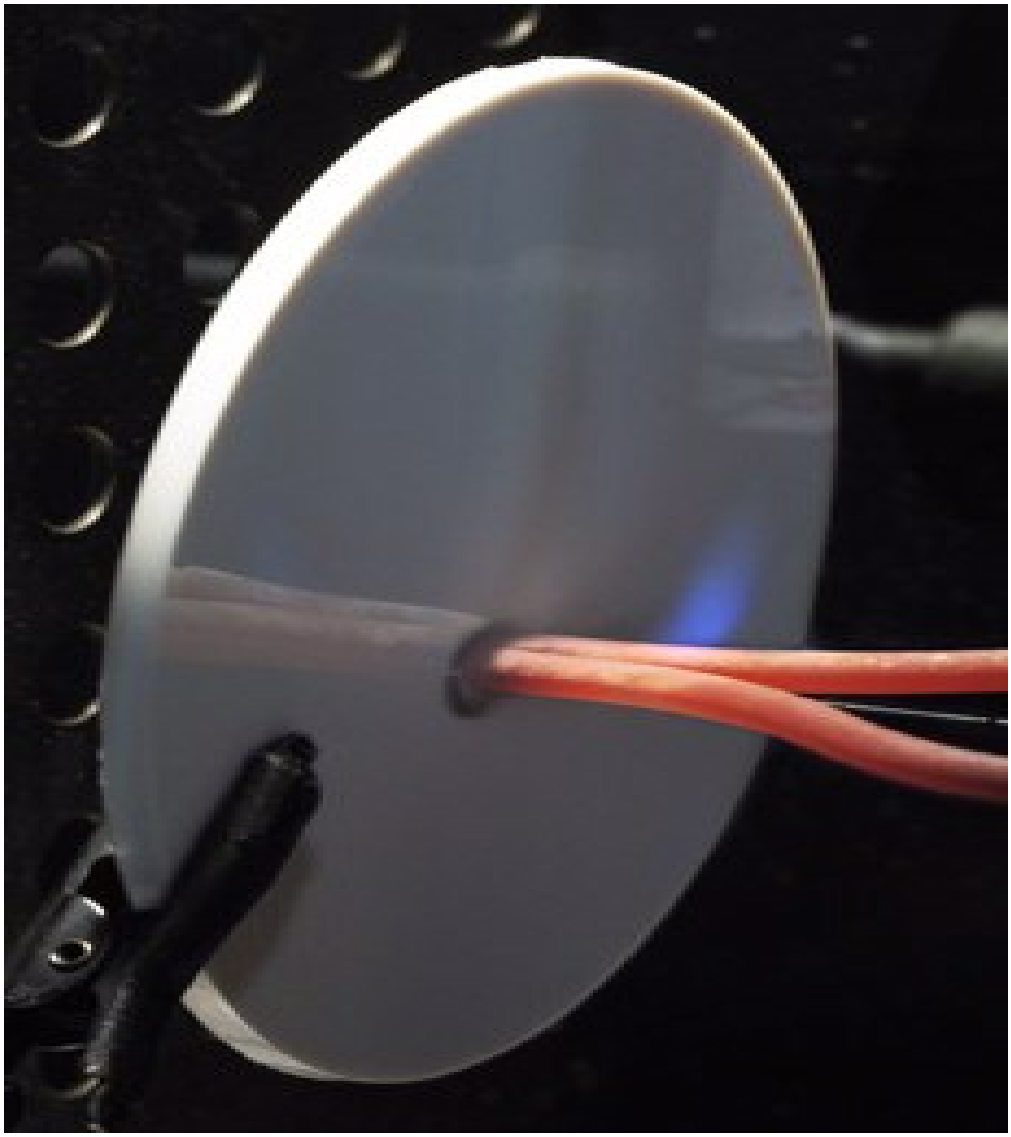
Picture 1: “Brightly shining flame”
Result: Test failed



Picture 2: Phenomenon of a “Blue Corona”
Result: Test OK



Picture 3: Occurrence of “Blue corona tailings over the glow wire”
Result: Test OK



Decision:

The “Corona-Effect” shall not to be taken into account.

The “Corona-Effect” is not considered as flame in the sense of the Glow Wire Test standards.